

Industry-Region Fit Research Summary

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Summary of the doctoral dissertation "Startup Success Drivers in Regional Ecosystems: Empirical Validation of the Industry-Region Fit Framework" (Dongguk University, 2025)

The question

Startups in the same industry often do well in one region and struggle in another. Regions put real money into changing that, through accelerators, university programs, cluster strategies and public funds. Which of these matters most is still an open question.

The Industry-Region Fit (IRF) framework looks at three parts of a regional ecosystem. Industry specialization is how concentrated a region is in particular industries. Network strength is how closely founders, investors and partners work with each other. Institutional support covers accelerators, incubators, university programs, government initiatives and venture capital infrastructure. The dissertation asks how each part relates to startup performance. It also asks whether institutional support changes how much the other two matter.

Data and design

The study covers 1,600 technology startups founded between 2015 and 2023 in Silicon Valley, New York, Boston and Los Angeles, with 400 firms in each region. I started with these four hubs because they have the richest public data and differ in how their ecosystems are built. Firm data come from Crunchbase, PitchBook and AngelList.

Performance is measured three ways. One is the funding stage a firm reaches. Another is how quickly it moves from one stage to the next. The third is how well it held up during the COVID-19 downturn, which the study treats as a natural experiment. The analysis uses region fixed effects, seemingly unrelated regression and hierarchical moderation tests, with robustness checks that include instrumental variables and propensity score matching.

What I found

- Institutional support had the strongest and most consistent association with startup performance ($\beta = 0.223$, $p < 0.001$).
- Network strength was also positively associated with performance, though more modestly. It mattered most for how quickly firms moved between funding stages.
- Industry specialization had a small effect on its own. It was clearly linked to success only in high-support settings ($r = 0.133$, $p < 0.001$). In low-support settings the correlation was 0.031 and not significant.
- Institutional support did not change the effect of network strength. Strong scores on all three parts together added no extra synergy ($\Delta R^2 = 0.00$), so the effects mostly add up.
- During 2020 and 2021, funding fell about 18% for startups in high-support regions and about 42% in low-support regions.
- The four hubs reached strong results through different mixes of the three parts.

What it means

For regions, the order of investment may matter. Support that helps founders reach capital and advice appears to come first. Specialization seems to pay off more once that support is in place. A region also does not need to copy

Silicon Valley. It can build on the mix of assets it already has.

These results are associations drawn from four large hubs. They point to where closer study would be most useful.

Next steps

The four hubs were a starting point. The next stage uses finer data and field work in regions that look quite different, including mid-sized and emerging ecosystems. I want to measure institutional support at the level of specific programs and follow how outcomes change as those programs start or grow. Working with local partners, I hope to turn the results into strategies that fit each region's industries and starting point. The longer aim is a model that explains differences across U.S. regions.

Related work

- Lee, T. (2025). *Startup success drivers in regional ecosystems: Empirical validation of the Industry-Region Fit framework* [Doctoral dissertation, Dongguk University].
- Lee, T., & Jeon, B. (2025). Rethinking startup success: Ecosystem acceleration and resource revaluation in post-pandemic ventures (2015-2023). *Asia-Pacific Journal of Business Venturing and Entrepreneurship*, 20(3), 1–20.
<https://doi.org/10.16972/apjbve.20.3.202506.1>
- Lee, T., & Jeon, B. (2025). Dynamics of industry-region fit: A simulation-based analysis of how entrepreneurial ecosystems influence startup performance. *Korean Journal of Entrepreneurship*, 20(3), 61–101 (in Korean).
<https://doi.org/10.24878/tkes.2025.20.3.61>